Sahaj Bhadja

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Summary

Infrastructure engineer with 2+ years building multi-tenant databases, distributed job systems, real-time services, and reliable data pipelines using Python, PostgreSQL, SQL Server, queues, cloud storage, concurrency, testing, and production debugging.

Education

University of Southern California, Los Angeles, CA **August 2024 – May 2026**
Master of Science in Computer Science *GPA: 3.75/4*
Coursework: Algorithms, Databases, Operating Systems, Web Technologies

Pandit Deendayal Energy University, Gandhinagar, India **November 2020 – May 2024**
Bachelor of Technology in Computer Science and Engineering *GPA: 3.88/4*
Coursework: Software Engineering, Operating Systems, Object-Oriented Programming, Data Retrieval

Experience

USC Dornsife **Los Angeles, CA**
Software Engineer *September 2025 – June 2026*
- Architected multi-tenant PostgreSQL and Azure SQL schemas for 500+ users with row-level isolation, normalized relationships, and indexed access patterns, reducing dashboard query latency from 1.2 seconds to 350 milliseconds.
- Built fault-tolerant FastAPI workers for 1,000+ monthly jobs with explicit states, retries, concurrency controls, and failure isolation, tripling throughput while preventing cascading failures across production workloads.

Grad8 **Los Angeles, CA**
Software Engineer Intern *June 2025 – August 2025*
- Engineered Python APIs and React workflows handling 10,000+ daily requests, integrating secure file ingestion, asynchronous evaluation, and persistent job histories while maintaining stable performance under peak load.
- Designed concurrent AWS RDS-backed processing for 1,000+ files per batch using multithreading and fault-tolerant execution, reducing turnaround by 60% while improving queue reliability and debugging visibility.

Aasma Technology Solutions **India**
Software Developer Engineer - (SecureSync) *August 2023 – July 2024*
- Developed cross-platform Python microservices and 20+ automated PyTest and Mockito suites through Jenkins CI/CD, improving release reliability while validating integrations across Windows, Linux, and macOS environments.
- Optimized REST API and ETL components using object-oriented patterns, AWS Glue, S3, and SQL, reducing response times by 30% and improving distributed data-pipeline processing efficiency by 68%.

Aasma Technology Solutions **India**
Full Stack Developer Intern - (VibrantChem) *May 2023 – August 2023*
- Built a MERN platform with REST APIs, WebSockets, and eight modular Node.js services, supporting 2M+ synchronized records while translating complex product requirements into maintainable real-time data workflows.
- Automated Docker-based testing and deployments through Jenkins CI/CD, reducing release time by 50% while improving code quality, documentation, observability, and defect resolution across frequent production updates.

LST Capitals **India**
Web Developer Intern *November 2022 – January 2023*
- Gathered client requirements and built Flask APIs for booking, scheduling, payment status, and user workflows across 50,000+ records, presenting system architecture and 12+ integration points before implementation.
- Integrated Firebase synchronization, JSON webhooks, and automated payment workflows, reducing manual processing by 70% while improving operational telemetry, troubleshooting visibility, and stakeholder communication around recurring production issues.

Skills

Systems: Python, C++, Java, JavaScript, TypeScript, SQL, Bash, data structures, algorithms, operating systems, object-oriented design, system design, concurrency, multithreading, scheduling, queues, backpressure, distributed systems; **Data & Services:** PostgreSQL, SQL Server, Azure SQL, MySQL, MongoDB, schema design, indexing, query optimization, row-level security, FastAPI, Flask, Node.js, REST APIs, WebSockets, microservices, ETL; **Infrastructure:** AWS, Azure, GCP, S3, AWS Glue, Docker, Kubernetes, Jenkins, Git, CI/CD, Linux, cloud storage; **Reliability:** PyTest, Mockito, unit testing, integration testing, debugging, instrumentation, monitoring, incident response, performance optimization, fault tolerance, data validation, technical documentation, Agile/Scrum

Projects

CPU-Scheduling Simulator Live | *C++, React.js, JavaScript, Algorithms*
- Implemented six scheduling algorithms with dynamic metrics and Gantt visualization, validating preemptive and non-preemptive execution across 1,000+ simulated process combinations while reducing algorithm-debugging time by 20%.

Big-Brain Live | *Python, FAISS, Pinecone, PostgreSQL, GCP*
- Built an AI document platform for 200+ users with secure uploads, vector indexing, semantic search, and collaborative workflows, maintaining 95% query accuracy across 50,000+ indexed documents.

FinNexus Live | *Next.js, TypeScript, MongoDB, WebSockets, Kubernetes*
- Built a scalable platform for 50,000+ users and 1M+ transactions, combining secure APIs, real-time synchronization, advanced filtering, and Kubernetes-backed deployments for reliable high-volume customer data workflows.